Mayo internal evaluation

Patient retrieval (all labels)

	Model	Overall (20)	Stage	Recurrence	Neuropathy	Lymphedema	pCR	MACE
Unimodal	*Notes_ModernBERT*	0.7336/0.7139	0.5400/0.4840	0.7400/0.7000	0.7500/0.8060	0.6200/0.5300	**0.2300/0.2140**	0.6100/0.5680
Unimodal	*Mammo_EfficientNet*	0.6926 /0.6939	0.4020/0.4080	0.6800/0.6740	0.8580/0.7500	**0.6600/0.5760**	0.1900/0.2300	0.5940/0.5840
Multimodal	*BCA*	0.6944/0.6974	0.3900/0.4080	0.6900/0.6720	**0.8800/0.8580**	0.5900/0.5780	0.2000/0.1700	0.5300/0.5980
Multimodal	*Time-awareBCA*	**0.7398/0.7165**	**0.5400/0.4900**	**0.7600/0.7120**	0.8600/0.8460	0.5100/0.5340	**0.2300/0.2140**	**0.6400/0.6000**

Multimodal risk prediction (1-5 year)

1. Patient Retrieval

		Overall			Risk=1 (Positive)			Risk=0 (Negative)		
Target	Method	P@1	P@5	P@10	P@1	P@5	P@10	P@1	P@5	P@10
1-year (261 pos / 208 neg)	Oracle Best	1.0000	1.0000	1.0000	–	–	–	–	–	–
	Random Baseline	0.5053	0.5053	0.5053	–	–	–	–	–	–
	Visual (pre-fusion)	0.5778	0.5382	0.5288	0.6398	0.5816	0.5705	0.5000	0.4837	0.4764
	Textual (pre-fusion)	0.6162	0.5774	0.5714	0.6398	0.6276	0.6257	0.5865	0.5144	0.5034
	Fused	**0.6375**	**0.6367**	**0.6360**	**0.6437**	**0.6529**	**0.6582**	**0.6298**	**0.6163**	**0.6082**
5-year (369 pos / 100 neg)	Oracle Best	1.0000	1.0000	1.0000	–	–	–	–	–	–
	Random Baseline	0.6638	0.6638	0.6638	–	–	–	–	–	–
	Visual (pre-fusion)	0.7655	0.7164	0.7075	0.8509	0.8114	0.8076	**0.4500**	0.3660	0.3380
	Textual (pre-fusion)	**0.7697**	0.7608	**0.7595**	**0.8618**	**0.8569**	0.8545	0.4300	0.4060	0.4090
	Fused	0.7612	**0.7625**	0.7520	0.8564	0.8542	**0.8457**	0.4100	**0.4240**	**0.4170**

2. Linear probe classification

Train/val/test=269/100/100

Target	Modality	Accuracy	Bal. Acc	AUROC
1-year (Train 269 / Test 100)	Visual (pre-fusion)	0.6000	0.5889	0.6102
	Textual (pre-fusion)	0.6600	0.6588	0.6865
	Fused	**0.6900**	**0.7063**	**0.7832**
5-year (Train 269 / Test 100)	Visual (pre-fusion)	0.6500	0.6553	0.6946
	Textual (pre-fusion)	**0.7800**	**0.8376**	**0.8965**
	Fused	0.6550	0.6990	0.8795